1. INTRODUCTION

1.1. PURPOSE

This document outlines the software and process requirements for the DGEOLOT project. The goal is to visualize parking lot geolocation data (latitude and longitude) using QGIS and publish the visualization to an internal server using IIS Manager for access by the Planner department.

1.2. SCOPE

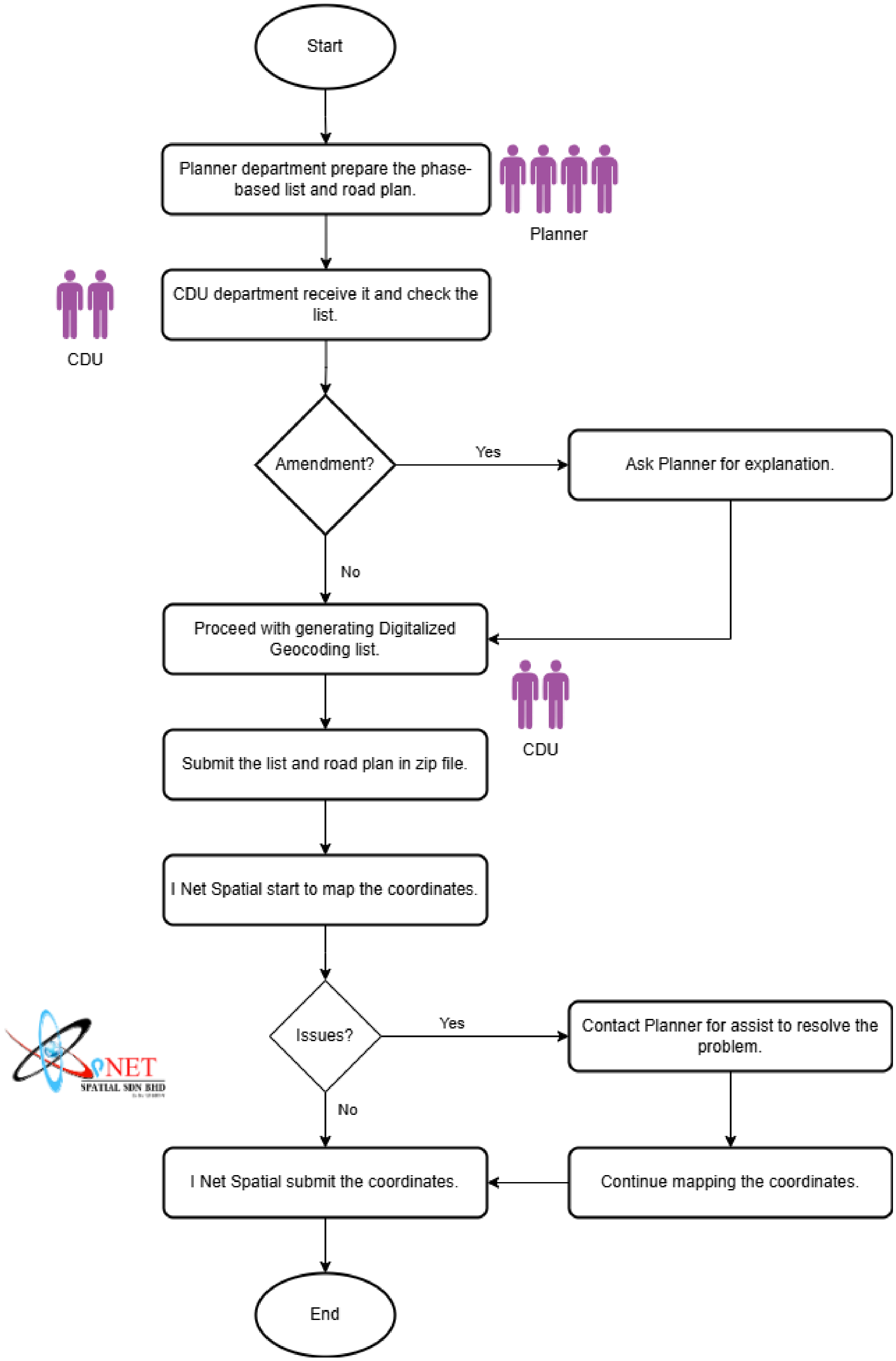
DGEOLOT is an internal GIS data visualization initiative. The system receives geolocation data from a consultant, visualizes it using QGIS by phase, and publishes the results to an internal web server for Planner review. There is no user-facing application; all tasks are carried out by departments using standard tools like QGIS and IIS Manager.

1.3. DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

Term	Definition
CDU	Corporate Development Unit
QGIS	Quantum Geographic Information System
SRS	Software Requirements Specification
DGEOLOT	Digitalized Geolocation
GIS	Geographic Information System
IIS	Internet Information Services

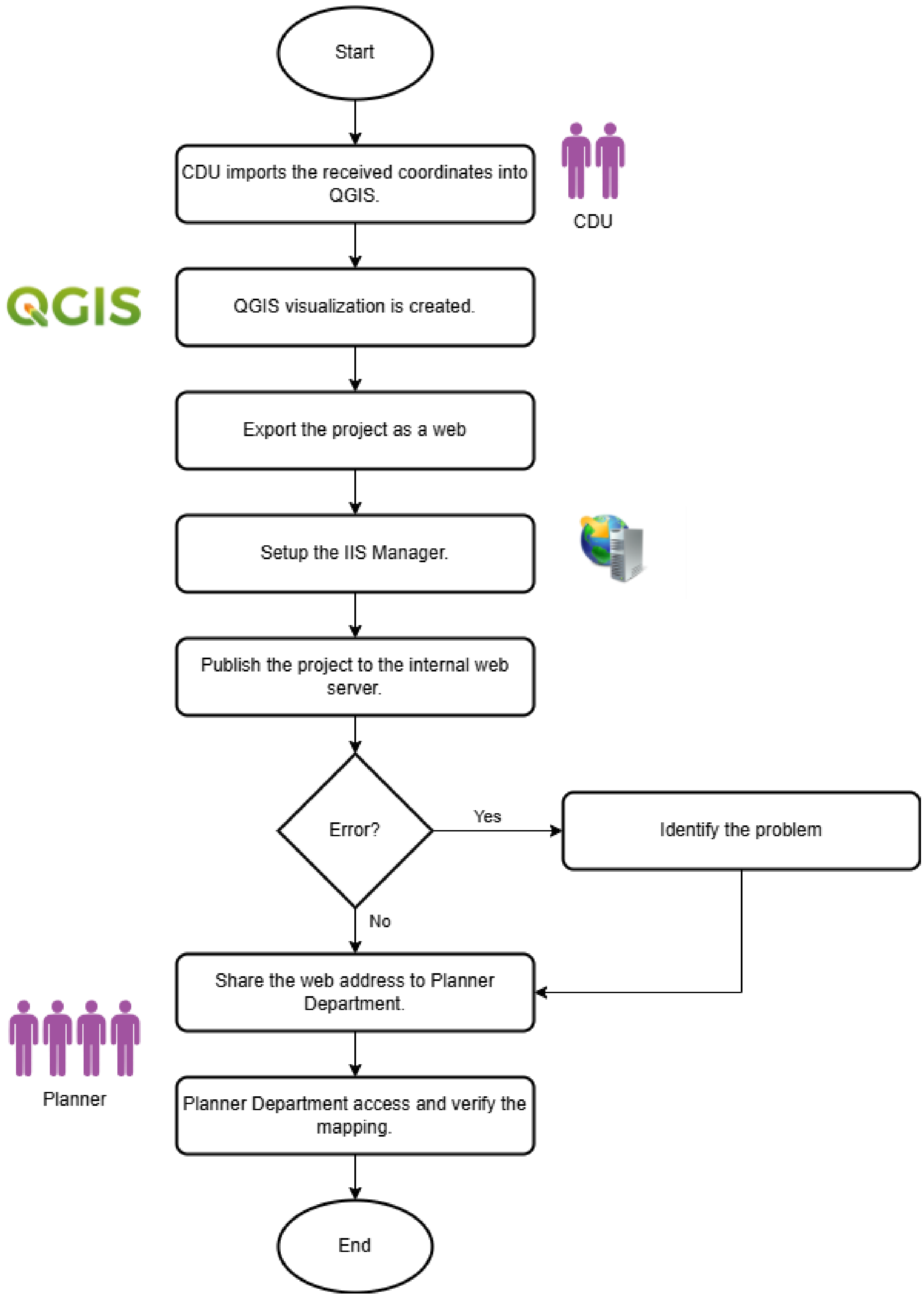
2. PROJECT WORKFLOW

2.1. DATA COLLECTION PROCESS



1) The Planner Department prepares a phase-based list and corresponding road plan of parking lots that require geolocation.
2) The CDU Department reviews the list to verify whether any roads are currently inactive or not in use.
3) If inactive roads are identified, the CDU Department requests justification from the Planner Department.
4) Once verified, the CDU Department proceeds to generate the Digitalized Geocoding List, including the “Kod Taman” and “Kod Jalan” for each parking lot.
5) The finalized list and road plan are compiled into a ZIP file and submitted to the consultant (I Net Spatial Sdn Bhd).
6) The consultant uses Google Maps to manually identify and record the latitude and longitude of each parking lot.
7) If the consultant encounters any issues, such as roads that cannot be found or matched, the Planner Department will be contacted to assist in resolving the problem.
8) Once complete, the geolocation data is returned to the CDU Department in a structured format, such as CSV or Excel.

2.2. VISUALIZATION AND PUBLISHING



1) The CDU Department imports the received coordinate data into QGIS.
2) A dedicated QGIS project is created for each phase, with visual markers representing the parking lot locations.
3) The project is then exported into a web-friendly format.
4) The IIS Manager is configured and set up on the internal server.
5) The QGIS project is published to the internal web server.
6) If any errors occur during publishing, the CDU Department will troubleshoot, identify the issue, and re-test the publishing method.
7) Once successfully published, the access URL is shared with the Planner Department.
8) The Planner Department accesses the provided address to review and validate the plotted parking lot locations.

3. FUNCTIONAL REQUIREMENTS

3.1. DATA INPUT

3.2. QGIS VISUALIZATION

3.3. DEPLOYMENT TO INTERNAL SERVER

3.4. ACCESS CONTROL

3.5. DATA MAINTENANCE

3.6. ERROR HANDLING

4. NON-FUNCTIONAL REQUIREMENTS

4.1. PERFORMANCE

4.2. AVAILABILITY

4.3. SECURITY

4.4. COMPATIBILITY

4.5. MAINTAINABILITY

- Support future project phases.
- Easy dataset replacement.
- Easy republishing.